

File

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 TERM 3

2013

**Mathematics**

**General Instructions**

- Working time – 75 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in Questions 6 to 13
- Start each question on a new page

**Total marks – 70 marks**

**Section 1 – 5 marks**

Attempt questions 1 -5

Allow about 7 minutes for this section

**Section 2 – 65 marks**

Attempt questions 6 to 13

Allow about 63 minutes for this section

Name: \_\_\_\_\_

Teacher: \_\_\_\_\_

**Section I (5 marks)**

Attempt questions 1 to 5.

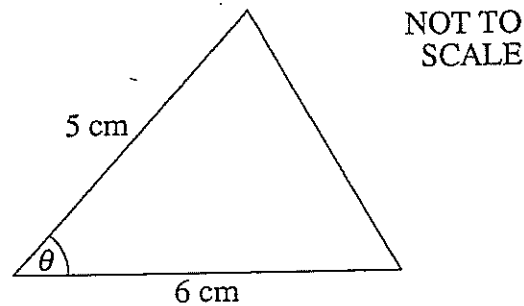
Allow approximately 7 minutes for this section.

Use the multiple choice answer sheet provided in your answer books.

Select the alternative A, B, C or D that best answers the question.

Fill in the response oval completely.

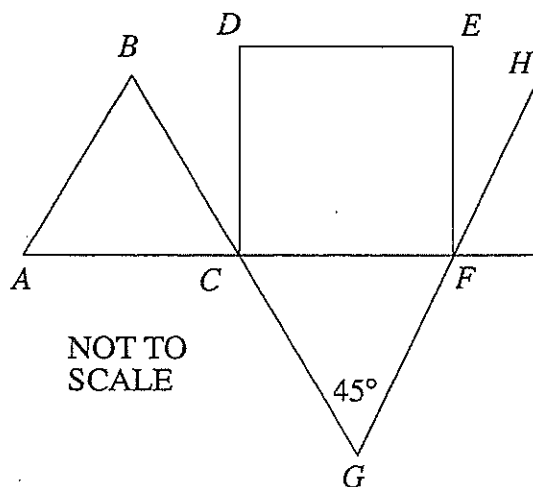
**Question 1**



Find the value of  $\sin \theta$  if the area of this triangle is  $10 \text{ cm}^2$ .

- (A)  $\frac{1}{3}$       (B)  $\frac{2}{3}$       (C)  $\frac{3}{2}$       (D) 3

**Question 2**



$ABC$  is an equilateral triangle.

$CDEF$  is a square.

$BCG$ ,  $ACF$ , and  $GFH$  are straight lines.

$\angle CGF = 45^\circ$ .

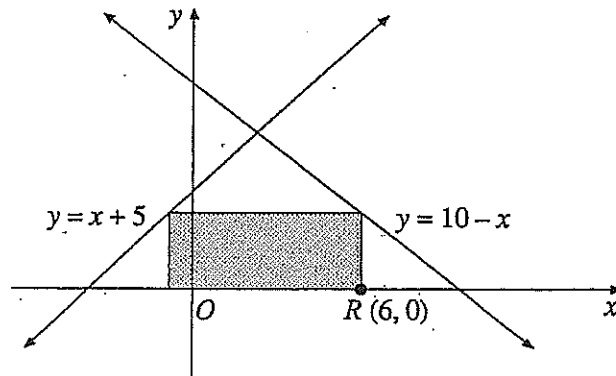
The size of  $\angle EFH$  is

- (A)  $15^\circ$   
(B)  $22\frac{1}{2}^\circ$   
(C)  $30^\circ$   
(D)  $45^\circ$

### Question 3

$R$  is the point with coordinates  $(6, 0)$ .

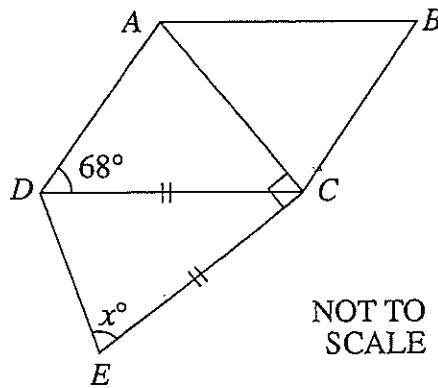
NOT TO SCALE



The area of the shaded rectangle in square units is

- (A) 20                      (B) 24                      (C) 28                      (D) 60

### Question 4



$ABCD$  is a rhombus.

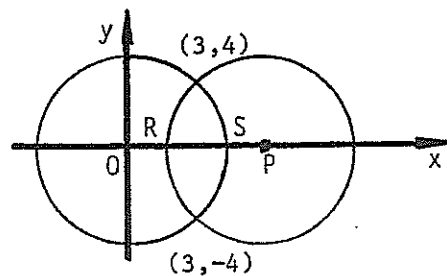
$DC = EC$ .

$\angle ADC = 68^\circ$  and  $\angle ACE = 90^\circ$ .

The value of  $x$  is

- (A) 22  
(B) 68  
(C) 73  
(D) 79

### Question 5



$O$  and  $P$  are centres of equal circles which intersect at  $(3, 4)$  and  $(3, -4)$ . The circles cut the  $x$  axis as shown at  $R$  and  $S$ .

$RS = ?$

A. 1

B. 2

C. 4

D. 5

**Section II - Attempt questions 6 – 13.**

**All answers and working to be done in your answer book.**

**Question 6 (8marks)**

- a) Simplify giving the answer to 2 significant figures

2

$$\frac{4.7 + \sqrt{3.2}}{1 + 4.15^2}$$

- b) Solve  $-3 \leq 1 - 2x < 5$

2

- c) If  $(2 + \sqrt{3})^2 = a + \sqrt{b}$  Find the value of a and b

2

- d) Solve  $x^2 \geq (x + 1)(x + 5)$

2

**Question 7 (8 marks) (Start a new page)**

- a) Factorise fully

i.  $8mn^2 + 6m^2n + 10mn$

1

ii.  $5x^2 - 33x + 18$

1

iii.  $125 - 27x^3$

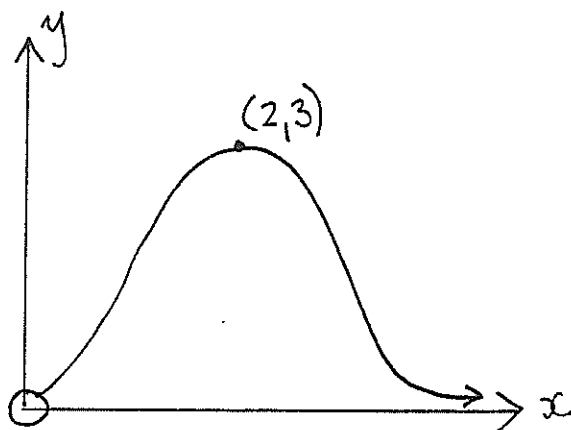
2

- b) If  $\sin \theta = \frac{4}{7}$  and  $\tan \theta < 0$  find the exact value of  $\cos \theta$

1

c)

3



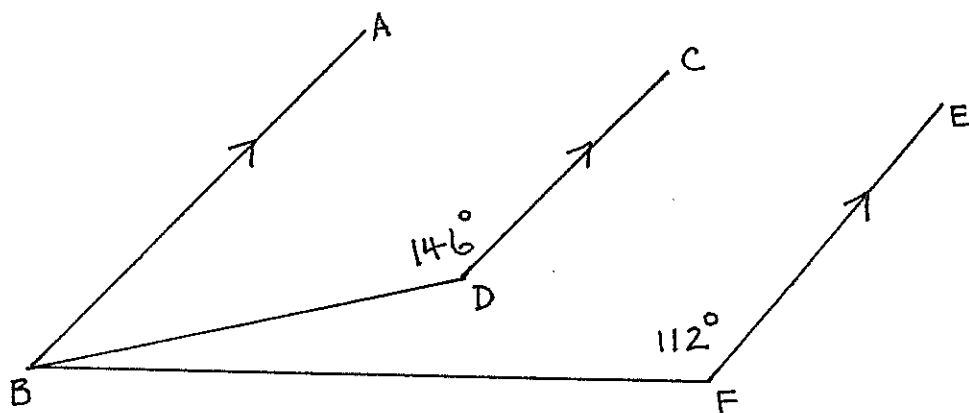
Consider the graph of  $y = f(x)$

- i. State its domain
- ii. State its range
- iii. Write down the equation (s) of any asymptotes

**Question 8 (9 marks) (Start a new page)**

a)

3



In the diagram above  $AB \parallel CD \parallel EF$ ,  $\angle BDC = 146^\circ$  and  $\angle EFB = 112^\circ$ . Show, giving reasons, that  $BD$  bisects  $\angle ABF$ .

b) Write  $\frac{1}{x+1} - \frac{1}{x-1}$  as a single fraction

2

c) Solve for  $x$

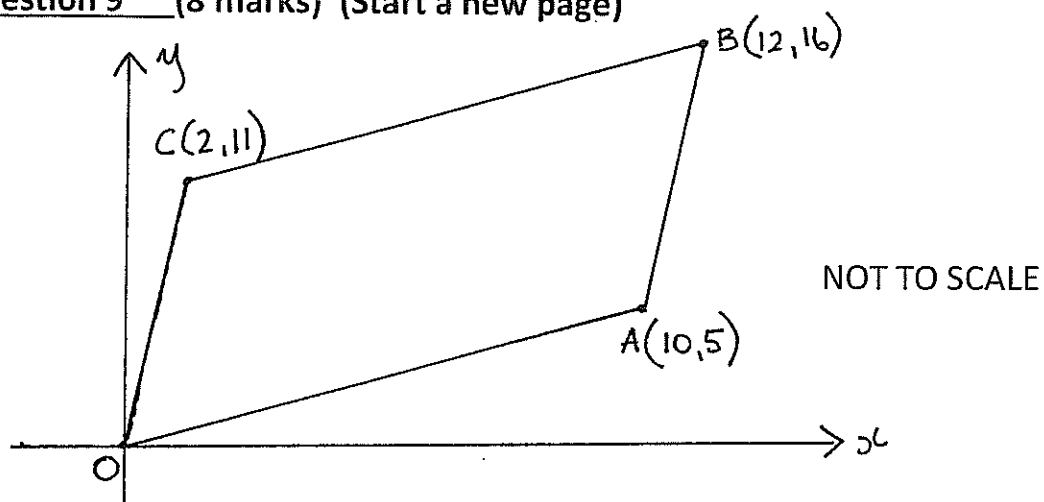
i.  $\frac{5}{8}(x+4) = 4x - \frac{1}{2}$

2

ii.  $x + \frac{2}{x} = 3$

2

**Question 9** (8 marks) (Start a new page)



In the diagram above A, B and C are the points (10,5), (12,16) and (2,11) respectively

Copy the diagram into your answer sheets

a) Find the distance AC

1

b) Find the midpoint of AC

1

c) Show the  $OB \perp AC$

2

d) Find the midpoint of OB and explain why OACB is a rhombus

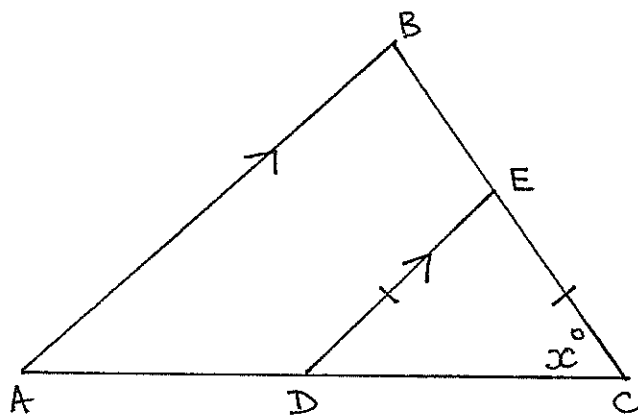
2

e) Hence find the area of OACB

2

**Question 10 (8 marks) (Start a new page)**

a)



In the diagram above  $AB \parallel DE$  and  $DE = EC$ . Let  $\angle ECD = x^\circ$

- i. Copy the diagram above onto your answer sheets
  - ii. Prove  $AB = BC$  ( reasons required) 3
- b)
- i. Sketch  $y = |x - 1|$  and  $y = |x|$  on the same axes (Label each graph carefully) 2
  - ii. Hence solve  $|x - 1| = |x|$  1
- c) Solve  $|x + 3| = 2x - 1$  2

**Question 11 (8 marks) (Start a new page)**

- a) Find the acute angle  $\theta$ , correct to the nearest minute,  
given that  $\tan \theta = 1 \frac{1}{3}$  2
- b) Simplify  $\frac{\cos (90 - \theta)}{\cos (180 - \theta)}$  2
- c) Solve  $2\sin \theta - 1 = 0$  for  $0 \leq \theta \leq 360^\circ$  2
- d) Simplify  $\sec^2 \theta - \tan^2 \theta$  2

**Question 12 (8 marks) (Start a new page)**

- a) Prove the identity 3

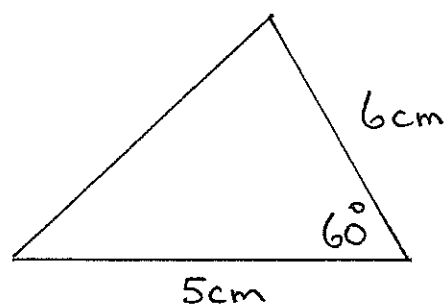
$$\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta} = 2\sec^2 \theta$$

- b) Solve the following trigonometric equations in the domain  $0^\circ \leq \theta \leq 360^\circ$
- i.  $3\cos^2 \theta + 5\cos \theta = 0$  3
- ii.  $\sqrt{3} \sin \theta - \cos \theta = 0$  2

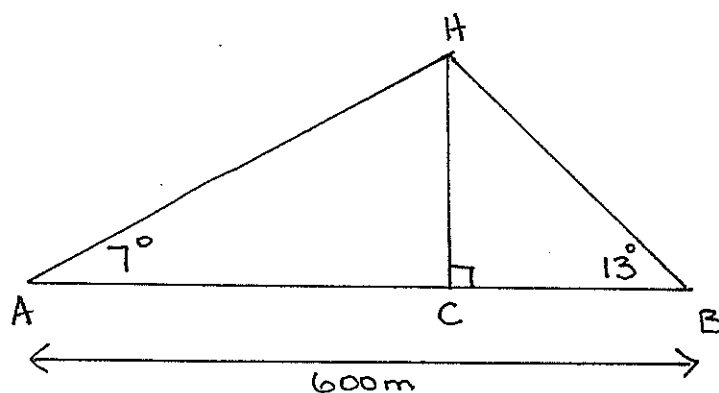
**Question 13 (8marks) (Start a new page)**

a) Find the exact area of the triangle below.

1



b) In the diagram below, a helicopter  $H$  is hovering above a straight, horizontal road  $AB$  of length 600 metres. The angles of elevation of  $H$  from  $A$  and  $B$  are  $7^\circ$  and  $13^\circ$  respectively. The point  $C$  lies on the road directly below  $H$ .



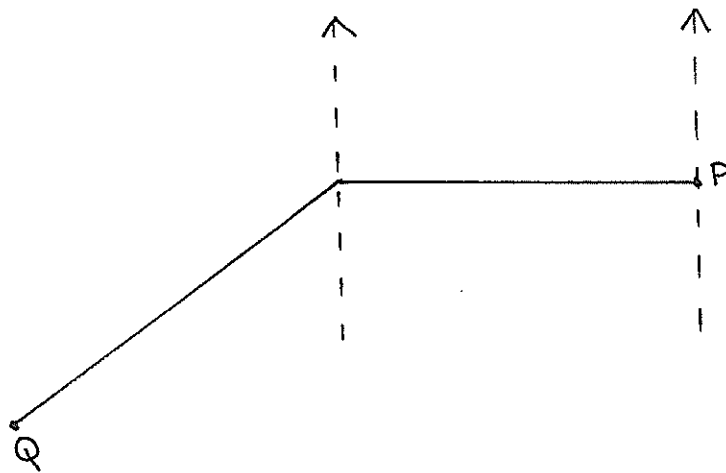
i. Use the sine rule to show that  $HB = \frac{600 \sin 7^\circ}{\sin 160^\circ}$

1

ii. Hence, find the height  $CH$  of the helicopter above the road, correct to the nearest metre.

1

- c) A ship sails 24 km west of port P and then 17 km on a bearing of  $S 42^{\circ}17' W$  to port Q.



- i. Copy the diagram above onto your answer sheets and show the information on it.
- ii. Calculate the distance of Q from P to one decimal place. 2
- iii. Calculate the TRUE bearing of the ship from P (to the nearest degree) 3



# MATHEMATICS

## SOLUTIONS YR 11 TERM 3 (WEEK 2) 2 UNIT

Question 1 B

Question 2 A

Question 3 C

Question 4 C

Question 5 C

Question 6

a) 0.36 (2 sig. fig)

b)  $-3 < 1 - 2x < 5$

$$-3 < 1 - 2x \quad 1 - 2x < 5$$

$$-2x \leq 4 \quad -4 < 2x$$

$$x \leq -2 \quad -2 < x$$

$$-2 < x \leq 2$$

c)  $(2 + \sqrt{5})^2 = a + \sqrt{b}$

$$4 + 4\sqrt{5} + 5 = a + \sqrt{b}$$

$$9 + 4\sqrt{5} = a + \sqrt{b}$$

$$a = 9 \quad b = 48$$

$$x^2 > (x+1)(x+5)$$

$$x^2 > x^2 + 6x + 5$$

$$-6x > 5$$

$$x < -5/6$$

Question 7

a) i)  $2mn(4n + 3m + 5)$

ii)  $(5x-3)(x-6)$

iii)  $(5-3x)(25+15x+9x^2)$

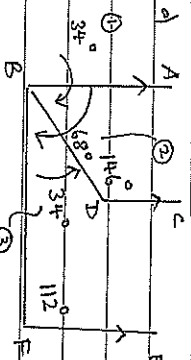
b)  $\frac{1}{\sqrt{33}} \quad \therefore \cos \theta = \frac{\sqrt{33}}{7}$

c) i) D.  $x > 0$

ii) R.  $0 < x \leq 3$

iii)  $y = 0$  (asymptote)

Question 8



$\angle ABD = 34^\circ$  (corresponding angles)

$AB \parallel CD$

$\angle BFC = 68^\circ$  ( $AC \parallel EF$ , corresponding angles)

$\therefore \angle BFE = 34^\circ$

$\therefore BD$  bisects  $\angle ABE$

b)  $\frac{1}{x+1} - \frac{1}{x-1} = \frac{x-1-(x+1)}{(x+1)(x-1)}$

$$= \frac{-2}{x^2-1}$$

c) i)  $\frac{5}{8}(x+4) = 4x - \frac{1}{2}$

$$5(x+4) = 32x - 4$$

$$5x + 20 = 32x - 4$$

$$24 = 27x$$

$$x = \frac{8}{9}$$

ii)  $x + \frac{2}{x} = 3$

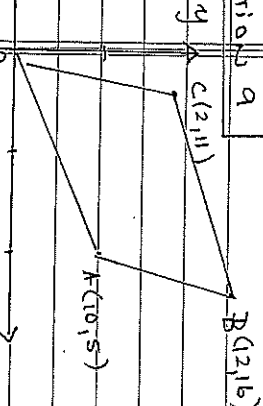
$$x^2 + 2 = 3x$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0$$

$$\therefore x = 2, x = 1$$

Question 9



a)  $AC = \sqrt{(10-2)^2 + (5-11)^2}$

$$= \sqrt{64 + 36}$$

$$= \sqrt{100}$$

$$AC = 10 \text{ units}$$

b) Midpt AC (6, 8)

c)  $m = \frac{16}{-8} = -2$

$$m = \frac{16}{-8} = -2$$

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$$\frac{4}{3} \times \frac{3}{4} = 1 \therefore OB \perp AC$$

$$\frac{4}{3} \times \frac{3}{4} = 1 \therefore OB \perp AC$$

$$\frac{4}{3} \times \frac{3}{4} = 1 \therefore OB \perp AC$$

d) Midpt OB (6, 8)

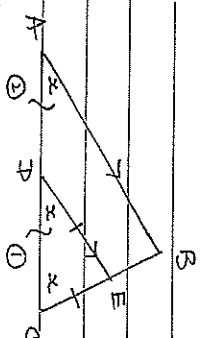
both diagonals bisected and meet at 90 degrees  $\therefore OABC \rightarrow$  rhombus

e)  $OB = \sqrt{11^2 + 16^2}$

$$OB = 20 \text{ units}$$

$$\therefore \text{Area } OABC = \frac{20 \times 10}{2} = 100 \text{ units}^2$$

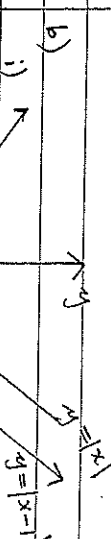
Question 10



$\angle EDC = \angle EAD$  (opposite equal sides in isosceles  $\triangle DEC$ )

$\angle BAD = \angle EDC$  (corresponding angles  $AB \parallel DE$ )

$\therefore AB = BC$  (opposite equal angles in isosceles  $\triangle ABC$ )



b) i)  $x = 1$

ii)  $x = \frac{1}{2}$

c)  $|x+3| = 2x-1$

$$x+3 = 2x-1 \quad x+3 = -(2x-1)$$

$$x = -4 \quad x+3 = -2x+1$$

$$3x = -2$$

$$x = -2/3$$

$$x = -2/3$$

$$x = -2/3$$

$$x = -2/3$$

$$x = -2/3$$

$$x = -2/3$$

$$x = -2/3$$

$$x = -2/3$$

$$x = -2/3$$

### Question 11

a)  $\tan \theta = \frac{1}{3}$   $\theta = 53^\circ 8'$   
(nearest min required)

b)  $\frac{\cos(90-\theta)}{\cos(180-\theta)} = \frac{\sin \theta}{-\cos \theta}$   
 $= -\tan \theta$

c)  $2\sin \theta - 1 = 0$   
 $\sin \theta = \frac{1}{2}$   $\checkmark$  S/A  $\checkmark$   
acute  $\theta = 30^\circ$   
 $\therefore \theta = 150^\circ, 30^\circ$

d)  $\frac{\sec^2 \theta - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{1}{1 + \tan^2 \theta}$

### Question 12

a)  $\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta} = 2\sec^2 \theta$

LHS =  $\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta}$   
 $= \frac{(1-\sin \theta) + (1+\sin \theta)}{1-\sin^2 \theta}$   
 $= \frac{2}{\cos^2 \theta}$   
 $= 2\sec^2 \theta$   
 $= \text{RHS}$

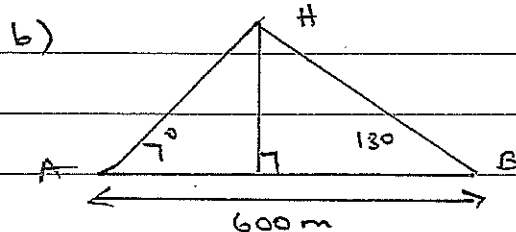
b) i)  $3\cos^2 \theta + 5\cos \theta = 0$   
 $\cos \theta (3\cos \theta + 5) = 0$   
 $\cos \theta = 0$   $\cos \theta = -5/3$   
 $\theta = 90^\circ, 270^\circ$  no solutions

ii)  $\sqrt{3}\sin \theta - \cos \theta = 0$   
 $\sqrt{3}\sin \theta = \cos \theta$   $\checkmark$  A  $\checkmark$  acute  $\theta = 30^\circ$   
 $\frac{\sqrt{3}\sin \theta}{\cos \theta} = 1$   $\checkmark$   $\therefore \theta = 30^\circ, 210^\circ$   
 $\therefore \tan \theta = \frac{1}{\sqrt{3}}$

### Question 13

a)  $A = \frac{1}{2} \cdot 6 \cdot 5 \cdot \frac{\sqrt{3}}{2}$

$A = \frac{15\sqrt{3}}{2} \text{ cm}^2$

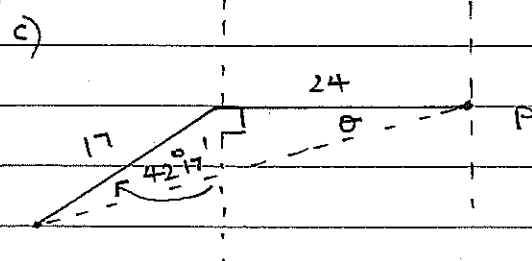


i)  $\frac{HB}{\sin 7^\circ} = \frac{600}{\sin 160^\circ}$

$HB = \frac{600 \sin 7^\circ}{\sin 160^\circ}$

ii)  $\sin 13^\circ = \frac{CH}{HB}$   
HB ← from above

$\therefore CH = HB \cdot \sin 13^\circ$   
 $= 48\text{m}$



i)  $QP^2 = 17^2 + 24^2 - 2 \times 17 \times 24 \times \cos(132^\circ 17')$   
 $\therefore QP = 37.6 \text{ km (1 dec pl)}$

ii)  $\frac{\sin \theta}{17} = \frac{\sin 132^\circ 17'}{QP}$

$\therefore \theta = 20^\circ$  (to nearest deg)

true bearing ship from P = 250° T